

Faithfulness of the representation on the CAR trace Hilbert space

MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation_injective

theorem

Proves that the representation $\tilde{\rho} : T \rightarrow B(H_\tau)$ is injective.

Statement

Let A be the completed CAR algebra and $T = C^*(\pi(A) \cup \{L_i : i \in I\}) \subseteq B(\mathcal{H})$ the algebra generated by the selected direct sum π of pure-state GNS representations indexed by their unitary-equivalence classes $i \in I$ and the unitaries L_i from the fixed homogeneity and shell construction. Write $\iota(a) = \pi(a)$ for the embedding $A \rightarrow T$. Let τ be the CAR trace and ϕ its chosen state extension to T . Write $\rho_\phi : T \rightarrow B(H_\phi)$ for the GNS representation of ϕ , and use the chosen unitary $E : H_\tau = L^2(A, \tau) \rightarrow H_\phi$ from the cited trace construction. For $\tilde{\rho}(t) = E^{-1}\rho_\phi(t)E$,

$$\tilde{\rho}(s) = \tilde{\rho}(t) \implies s = t \quad (s, t \in T).$$

Assumptions

The automorphisms and unitaries defining T are those fixed by CAR homogeneity. The cited injectivity theorem for the conjugated GNS representation applies to every such shell family, hence to this one.

Conclusion

Thus $\tilde{\rho} : T \rightarrow B(H_\tau)$ is faithful. Together with the separately established separability of H_τ , it realizes this same T faithfully on a separable Hilbert space.

This statement concerns injectivity of a representation, not faithfulness of a vector state, and does not assert irreducibility.

Proof route

Cancel the fixed unitary conjugation and use faithfulness of the GNS representation ρ_ϕ of the state extension.

▼ Proof steps

1. If $\tilde{\rho}(s) = \tilde{\rho}(t)$, then

$$\rho_\phi(s) = E\tilde{\rho}(s)E^{-1} = E\tilde{\rho}(t)E^{-1} = \rho_\phi(t).$$

The cited injectivity theorem for ρ_ϕ gives $s = t$. The Lean proof specializes this same argument to the fixed family.

Main citations

- The stated existence or structural result —
MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation_injective
- The exact representation being tested —
MathlibAnnex.CStarAlgebra.CAR.separableCounterexampleRepresentation
- Injectivity under the fixed unitary conjugation —
MathlibAnnex.CStarAlgebra.CAR.traceModelRepresentation_injective
- Faithfulness of the GNS representation of the extended state —
MathlibAnnex.CStarAlgebra.CAR.tracialRepresentation_injective
- Separability of the fixed Hilbert space —
MathlibAnnex.CStarAlgebra.CAR.separableSpace_separableCounterexampleHilbertSpace

Supporting route explanation

Lean source signature (exact)

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theorem separableCounterexampleRepresentation_injective :  
  Function.Injective separableCounterexampleRepresentation
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In the source	Mathematical meaning
AtomicCounterexampleAlgebra; separableCounterexampleRepresentation	The same fixed shell-generated algebra T and its representation $\tilde{\rho} : T \rightarrow B(H_\tau)$, where $H_\tau = L^2(A, \tau)$ is the CAR trace GNS space.
Function.Injective separableCounterexampleRepresentation	For all $s, t \in T$, equality $\tilde{\rho}(s) = \tilde{\rho}(t)$ as bounded operators implies $s = t$. Thus this action is faithful. The chosen $E : H_\tau \rightarrow H_\phi$ and ρ_ϕ in its defining conjugation are not new inputs; irreducibility is not asserted.

Further source notes: Its proof calls traceModelRepresentation_injective homogeneityShellFamily; the unitary E and ρ_ϕ are in that linked construction, not additional assumptions in this signature.

Earlier published Card and PDF

▼ Exact content identity

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